System Architectures

Overview

- Architecture defined
- Developing architectures
- Types of architectures
- Generic Web Architecture
- Layered-aspect architectures
- Data-aspect architectures
- Example

A Definition

Architecture

The embodiment of concept, and the allocation of physical/informational function (process) to elements of form (objects) and definition of structural interfaces among the objects

- 5 key attributes of software architectures

  Structure, Elements, Relationships

  Analysis => Implementation

  Multiple viewpoints (conceptual, runtime, process & implementation)

  Understandable

  Framework for flexibility

http://www.sei.cmu.edu/architecture/definitions.html

System Architecture

- System architecture, which translates the logical design of an information system into a physical blueprint or architecture.
- An information system requires hardware, software, data, procedures and people to accomplish a specific set of functions.
- An effective system combines those elements into an architecture.
- System architecture translate the logical design of an information system into physical structure that includes hardware, software, network support, processing methods and security.

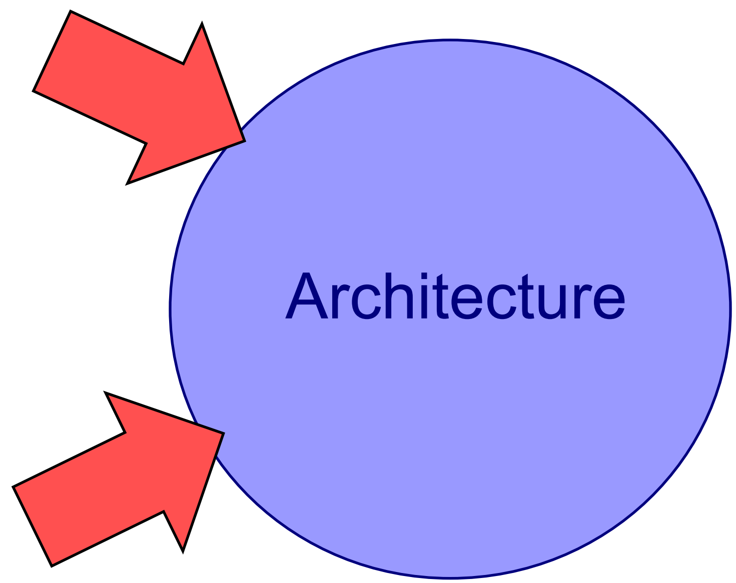
Planning the architecture

- Every information systems involves three main functions:
  - Data storage and access methods
  - Applications program to handle the processing logic
  - Interface that allows users to interact with the system.
- System Architecture: Defining the overall structure of the system, including major components and their interactions.
- Technology Selection: Deciding on the technologies, tools, and platforms to be used.

Developing Architectures

- Influences on Architectures

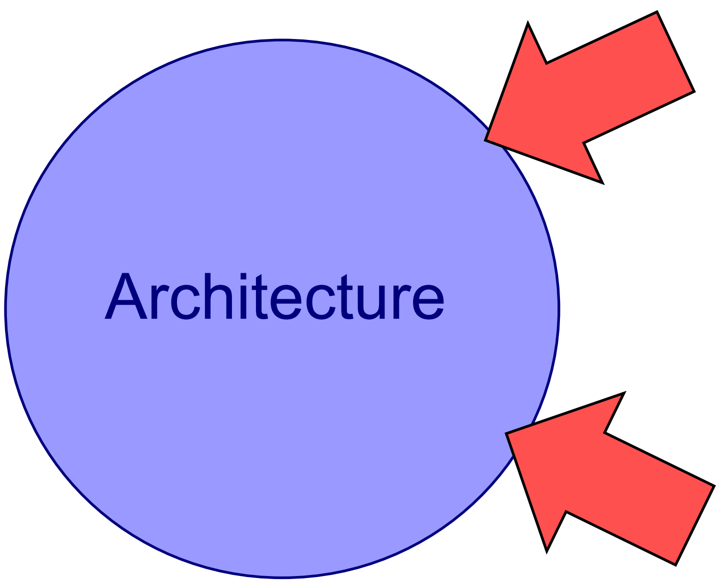
<u>Functional Requirements</u>
- Clients
- Users
- Other Stakeholders



<u>Experience with</u>
- Existing Architecture
- Patterns
- Project Management
- Other?

Developing Architectures

- Influences on Architectures (continued)



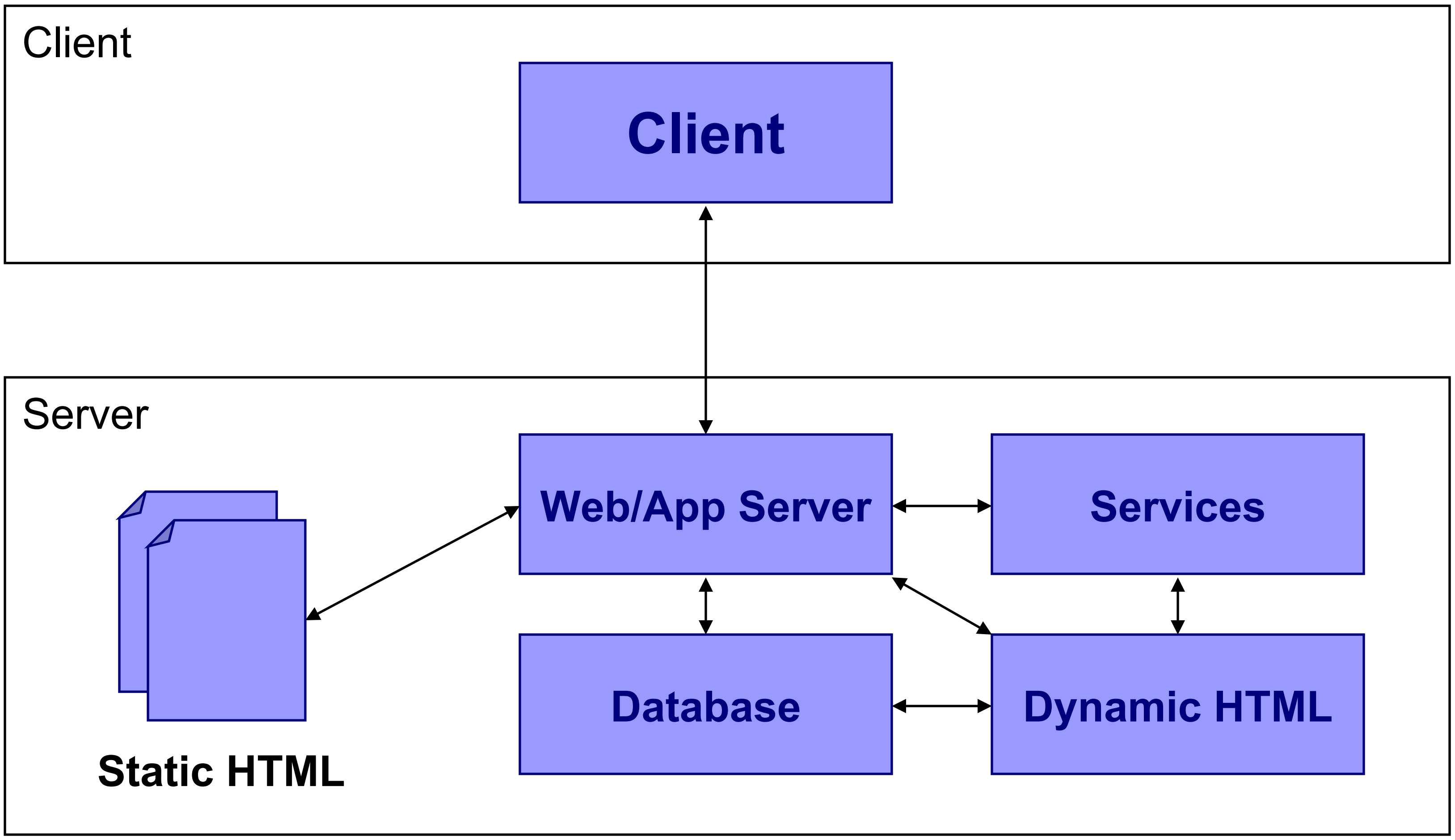
__Quality considerations with__

- Performance
- Scalability
- Reusability
- Other?

__Technical Aspects__

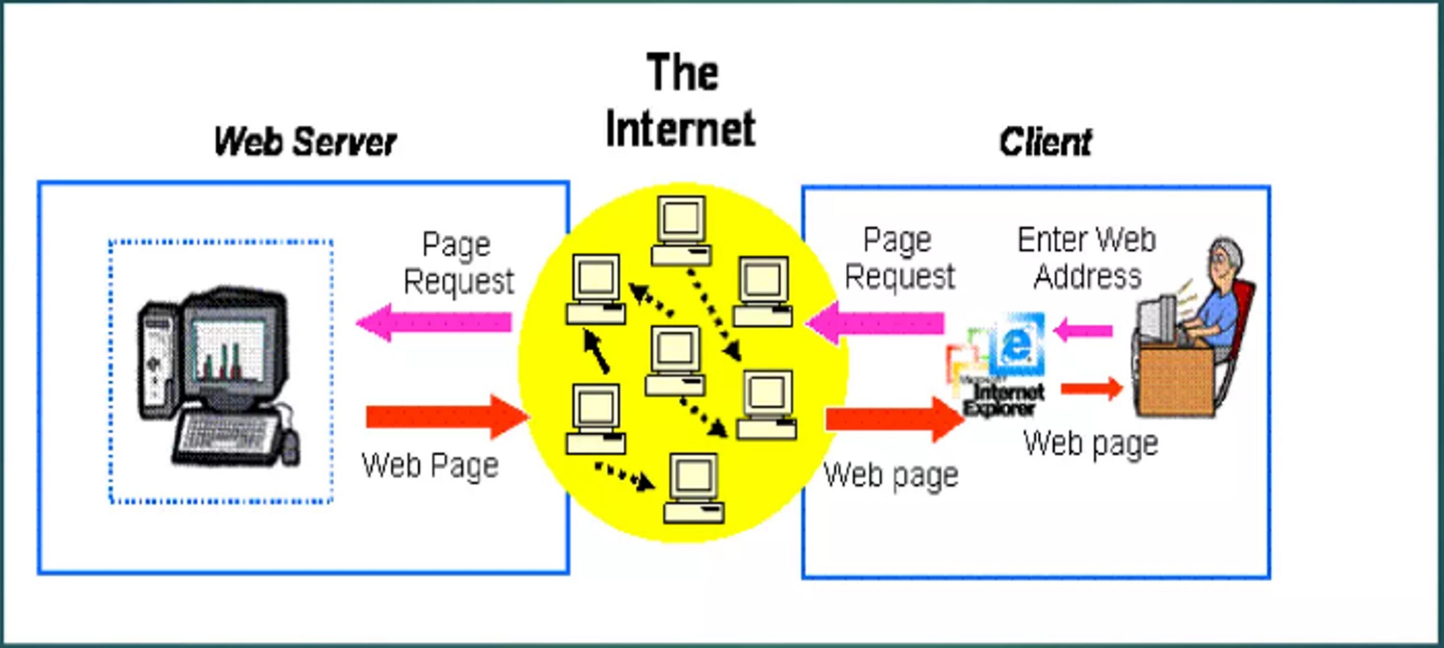
- Operating System
- Middleware
- Legacy Systems
- Other?

Client/Server (2-Layer)

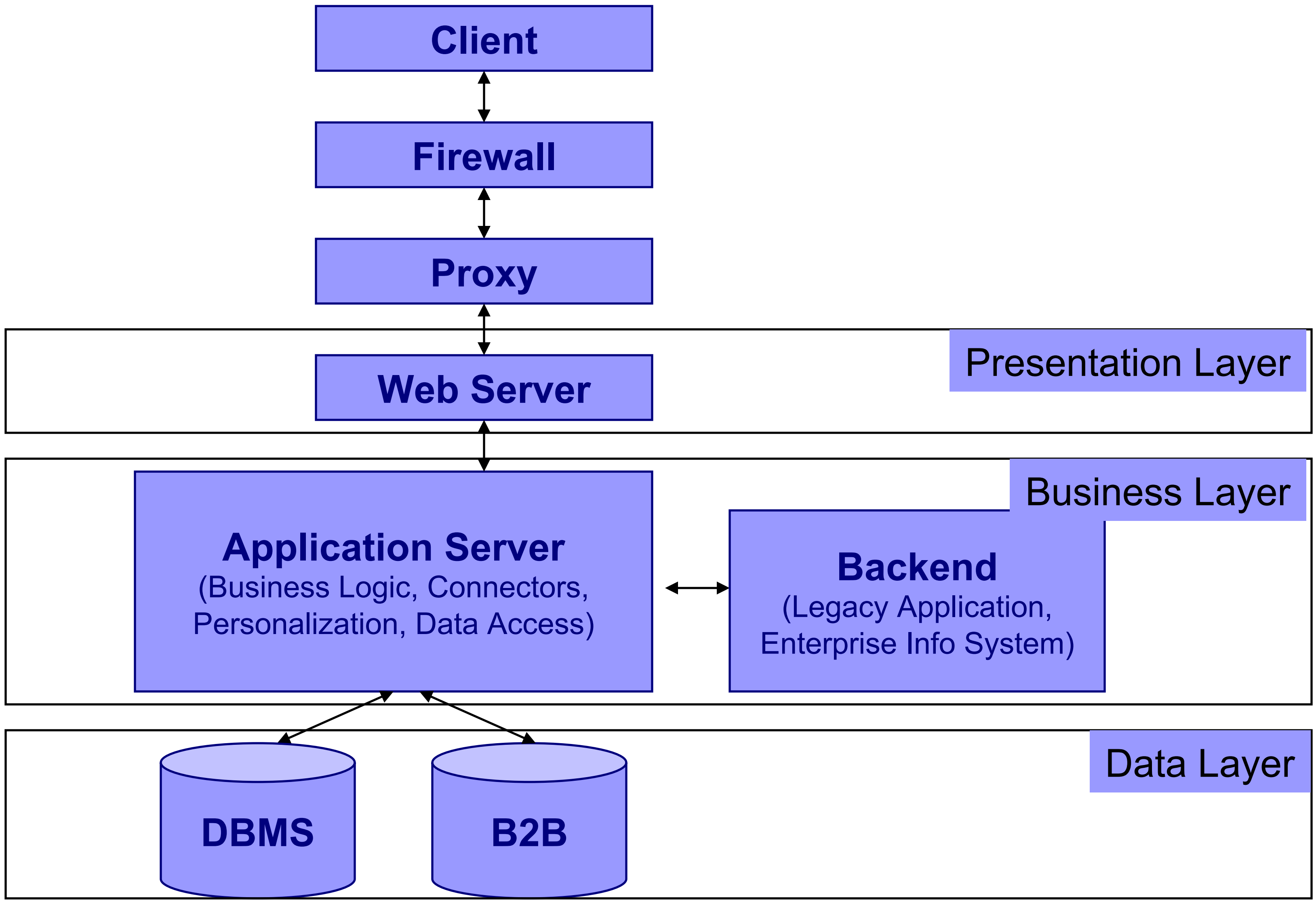


2 Tier Architecture

- This simple scenario, where the Web server is connected to one or more clients is known as a 2 tier architecture model.
- **Figure below demonstrates how Web pages are accessed via a browser, using a 2 tier architecture.**



N-Layer Architectures



Why an N-Layer Architecture?

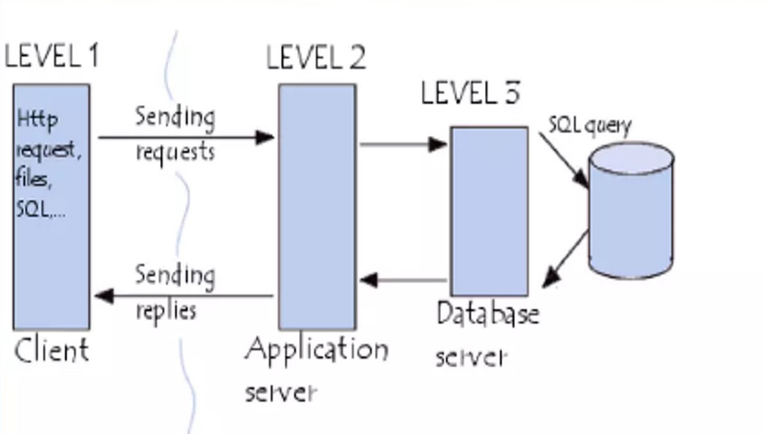
- Separating services in business layer promotes re-use different applications
  - Loose-coupling – changes reduce impact on overall system.
  - More maintainable (in terms of code)
  - More extensible (modular)
- Trade-offs
  - Needless complexity
  - More points of failure

3 Tier Architecture

3-Tier Architecture

- Generally computing applications consist of three different and distinct types of functionalities.

- **Presentation Services**
- **Functional logic**
- **Data Management**



More on Proxies

- Originally for *caching* data
- Can also server other roles:

  Link Proxy

  - *Persistent URL's* – maps the URL the client sees to the actual URL.
  - *AJAX* – allows data from a 2nd server to be accessed via a client script.

  History Proxy

  - HTTP is stateless - navigation history cannot be shared across multiple websites.
  - Multiple companies can access a server-side cookie (e.g. DoubleClick)

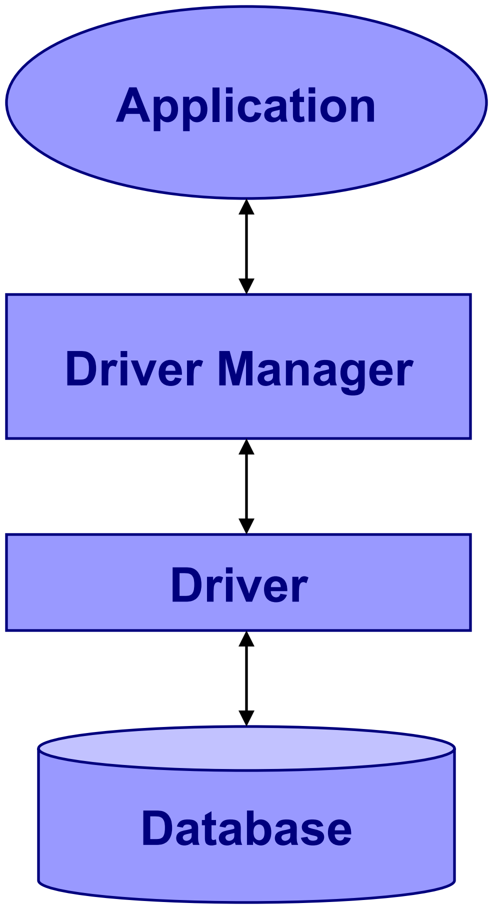
Integration Architectures

- Enterprise Application Integration (EAI)
- Web Services
- Portals/Portlets
- Challenges/Pitfalls
    - Cannot separate logic & data in legacy systems
    - Incompatible schemes
    - Poor documentation
    - Measuring performance/scalability

Data-Aspect Architectures

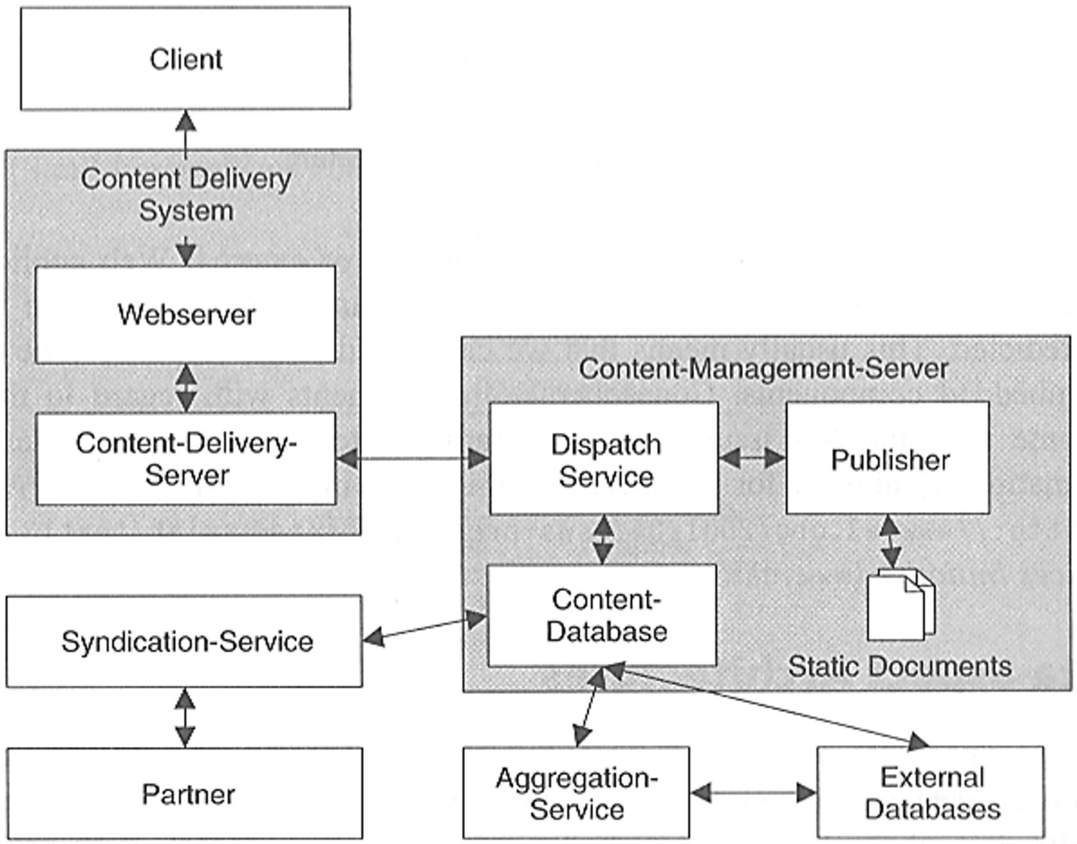
- Data can be grouped into either of 3 architectural categories:
  1. Structured data of the kind stored in DBs
  2. Documents of the kind stored in document management systems
  3. Multimedia data of the kind stored in media servers

- Structured data (JDBC/ODBC)
  - Accessed either directly via a web extension (for 2-tier) or over app server (for n-tier).
  - Since DB technology are highly mature, they are easy to integrate
  - Easy to implement
    - APIs are available to access DBs (e.g., JDBC, ODBC)



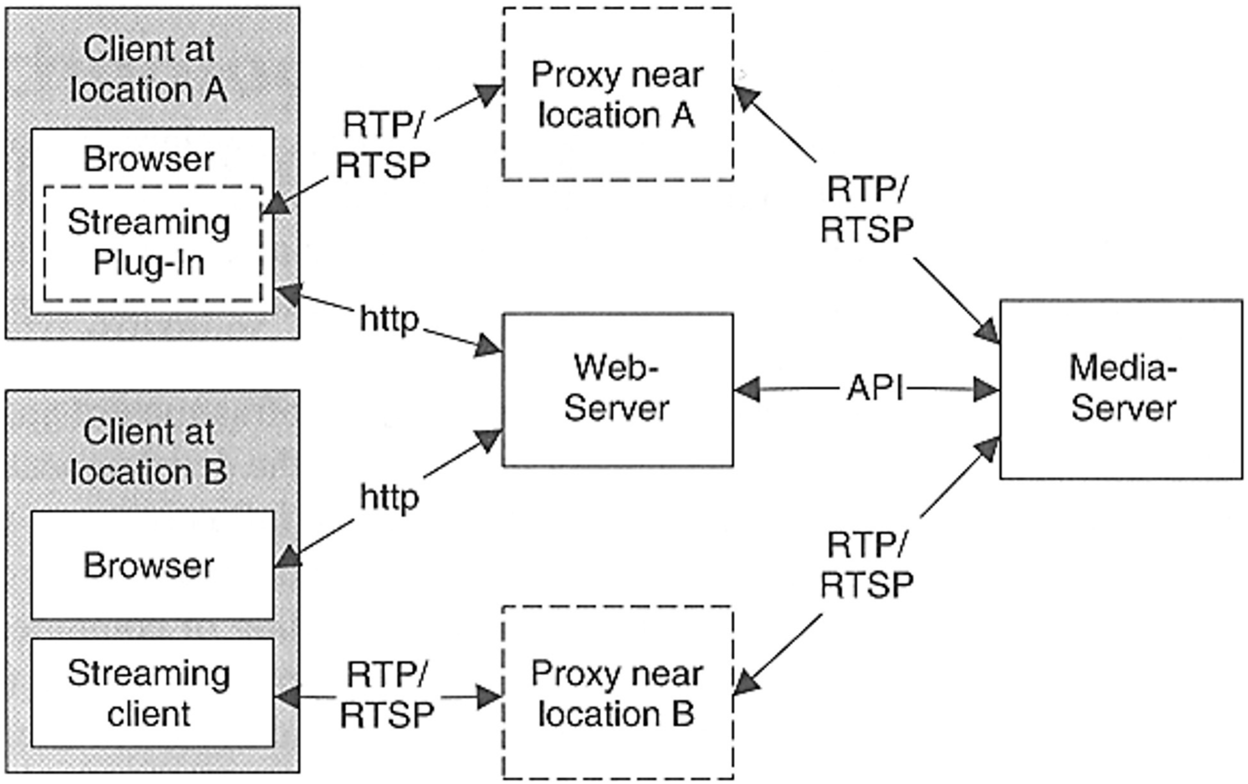
Data-Aspect Architectures

- Web Document Management



Data-Aspect Architectures

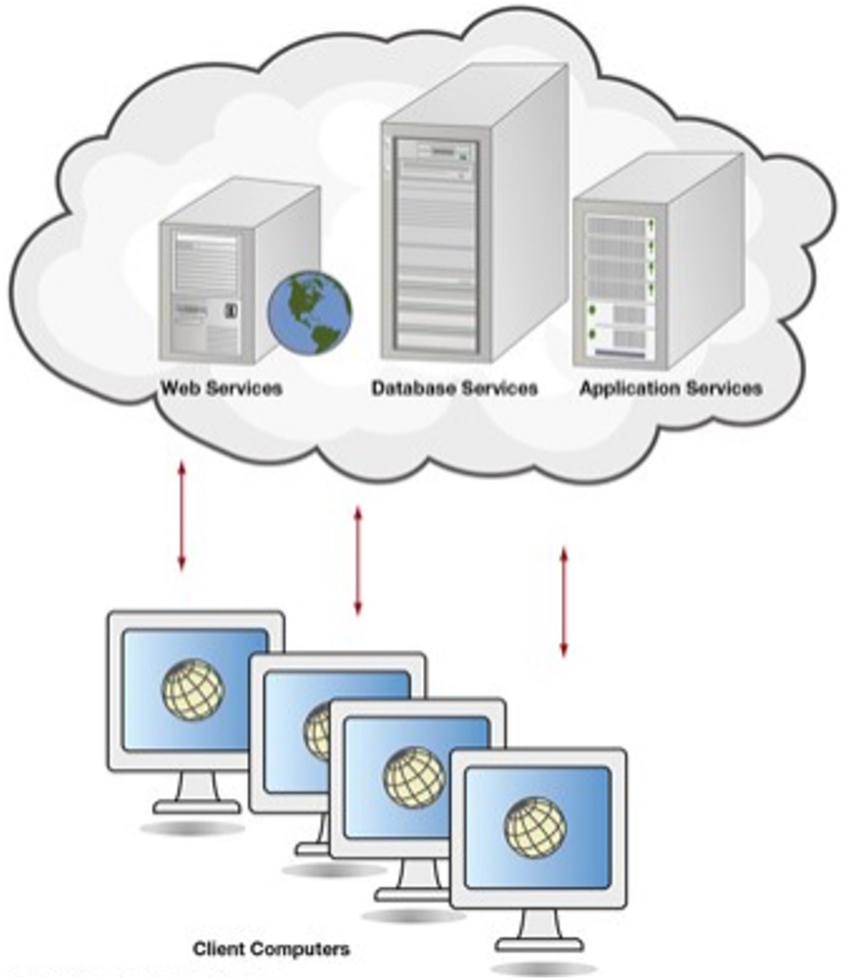
- Web Multimedia Management: Point-to-point



Cloud Computing

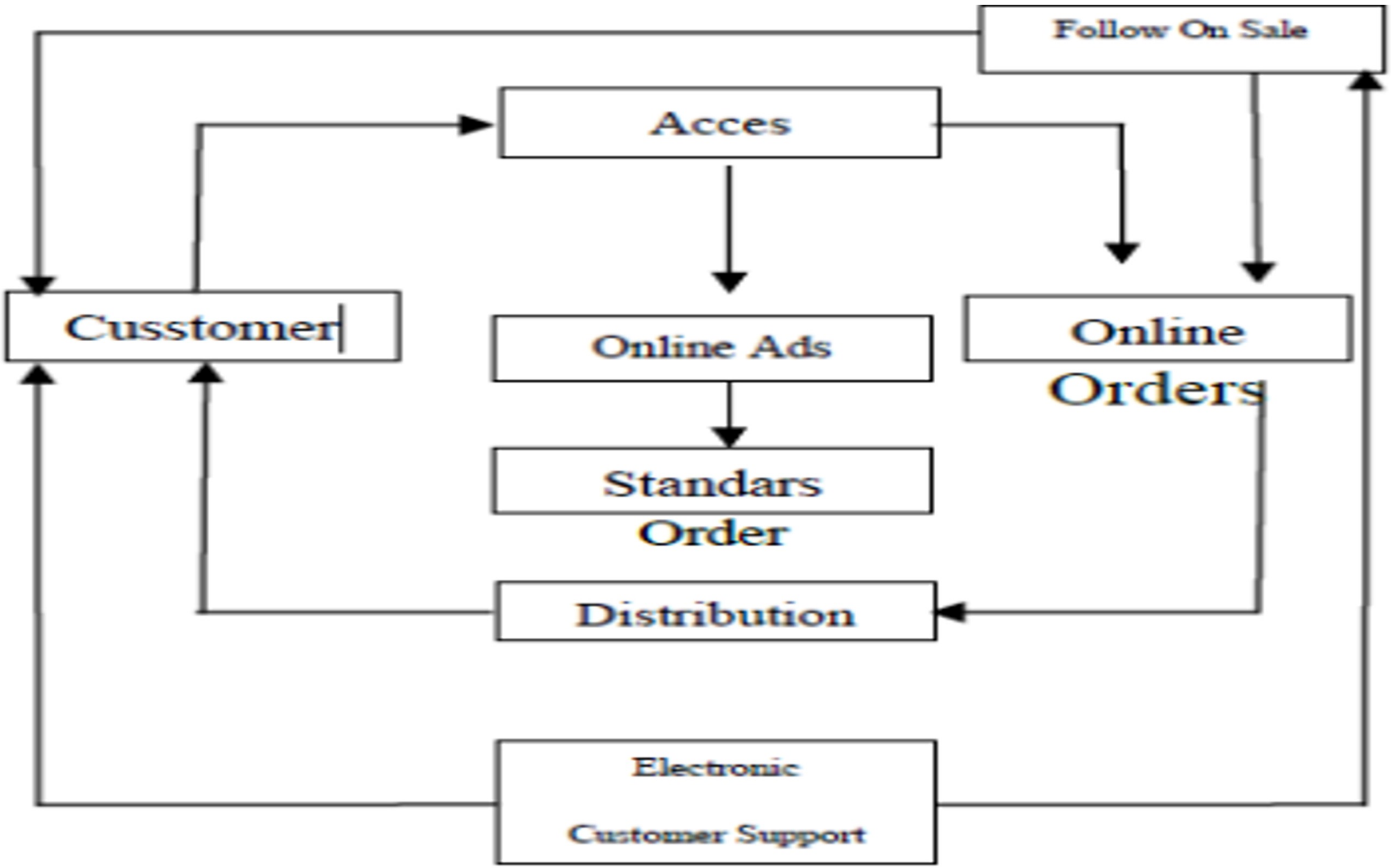
- Uses web services, database services, and applications over the internet without the investment of corporate or personal hardware, software, or software tools
- These “virtualized resources” have the ability to grow and adapt to changing business needs making them scalable to suit growing or changing demand by users
- Shared IT resources allow for cost sharing
- Improved disaster recovery due to redundant sites

Figure 1.6 Cloud Computing Offers Many Services



Example

Designing Consultant Services Sales System through Online Store



Soegoto, E. S., Juliana, R., & Oktafiani, D. (2018, August). Designing Consultant Services Sales System through Online Store. In *IOP Conference Series: Materials Science and Engineering* (Vol. 407, No. 1, p. 012027). IOP Publishing.

A system architecture for intelligent browsing on the web

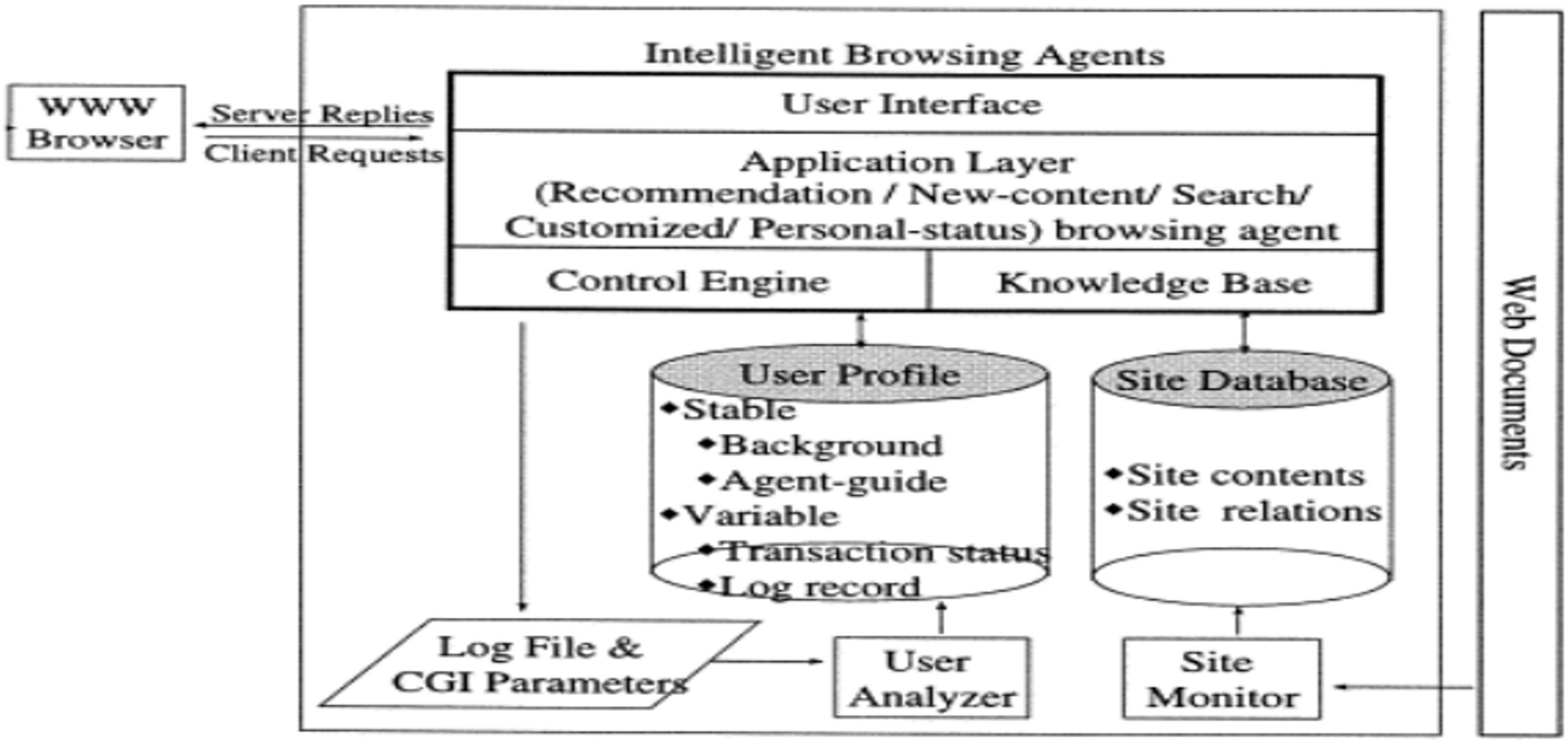


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Fig. 1. A system architecture for intelligent browsing.

Lai, H., & Yang, T. C. (2000). A system architecture for intelligent browsing on the web. *Decision Support Systems*, *28*(3), 219-239.